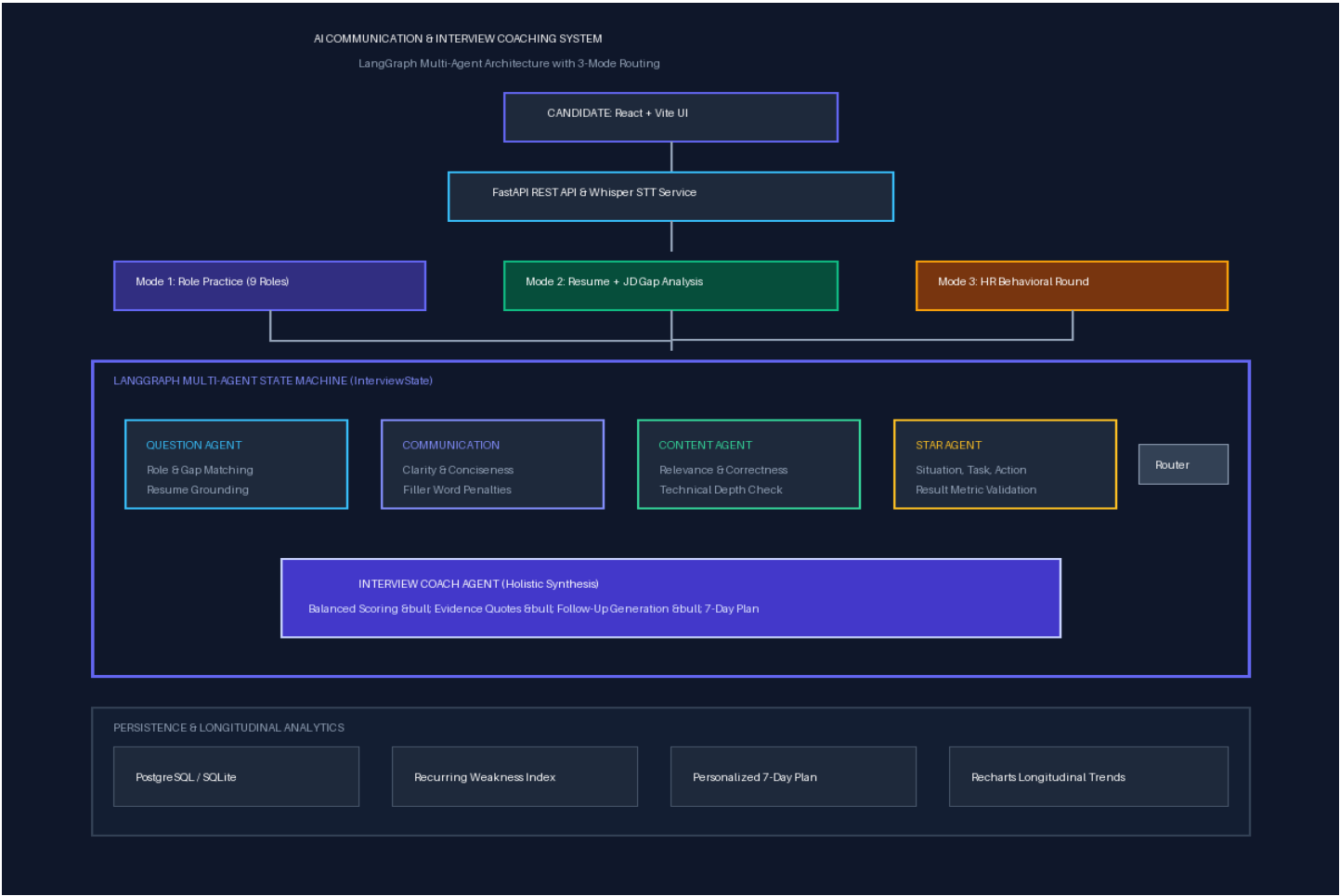


AI-Powered Communication & Interview Coaching System

Comprehensive System Architecture & Multi-Agent Design Document



1. Architectural Overview & Workflow

The system utilizes LangGraph as a stateful, compiled multi-agent orchestration engine. Unlike generic single-prompt chatbots, responses are evaluated across specialized orthogonal dimensions: Communication clarity, technical content depth, and STAR behavioral structure. A conditional router triggers deeper specialist diagnostics if any core dimension scores below threshold, after which the Coach Agent performs holistic evidence-grounded synthesis.

2. Three Core Interview Entry Modes

Mode	Description	Evaluation Focus
Mode 1: Role Practice	Select from 9 curated preset roles (SDE, DevOps, AI/ML, PM, Sales, Data, etc.)	Domain concepts, system design, role-specific competencies.
Mode 2: Resume + JD	Upload resume (PDF/TXT) and paste Job Description for automated gap analysis.	Resume-grounding; probing undemonstrated JD skills.
Mode 3: HR Round	Curated behavioral tracks (Conflict, Feedback, Stress, Failure, Motivation, etc.)	Interpersonal communication, radical empathy, STAR structure.

3. Scoring Rubric & Longitudinal Progress

Overall Score Equation: Relevance (25%) + Content Quality (25%) + Communication (20%) + Structure (15%) + Completeness (15%). Cross-session intelligence detects recurring gaps (appearing in ≥ 2 sessions) and generates an adaptive 7-Day Personalized Improvement Plan.